SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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ITA0334- MOBILE COMPUTING
ASSIGNMENT 2 – CO2

CO2 Coverage: Mobile Telecommunication Systems; Multiple Access Techniques (SDMA, TDMA, FDMA, CDMA, OFDMA); GSM Architecture, Services, Radio Interfaces, Localization, Calling, Handover & Security; Advanced GSM Extensions (GPRS, HSCSD, DECT).

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Nature of Assignment : Individual

Duration : 1 Week

Total Marks : 50

Submission Date : 13-11-2026

Q1 – Mobile Communication or Mobile Computing? **5 Marks**

Problem Context – Field-Service Technology Audit

A utility company equips its field staff with: (a) a ruggedised tablet fetching work orders over cellular data, (b) a laptop plugged into Ethernet at different offices, (c) a fixed wireless router at a remote substation, (d) a push-to-talk handset, and (e) a tablet app that records meter readings offline and synchronises later.

Your Challenge

Investigate whether “wireless” and “mobile” mean the same thing, and whether mobile communication and mobile computing are the same thing.

Before analysis:

- Define at least four of your own classification criteria (e.g., user mobility, wireless link, tolerance of disconnection, location/context awareness, device resource limits).
- Predict how each of the five cases is classified (fixed-wired, nomadic, fixed-wireless, mobile) and whether it is mainly a mobile-communication or a mobile-computing problem, giving reasons.
- Design one additional ambiguous case of your own to test your criteria.

Test Your Reasoning

Build a small simulation of case (e) with link availability of 90%, 60% and 30%. Compare two strategies: send immediately (fail if disconnected) vs store-and-forward queue. Record updates delivered, delayed and lost, and mean delivery delay.

Inquiry

Which characteristics of mobile computing are exposed by your results that a wireless link alone does not solve? Which device categories need application-level changes?

Evidence Expected: Own classification criteria, reasoned predictions, an engineered ambiguous case, simulation counts for both strategies, and a justified conclusion.

Answer:

1. Aim and problem interpretation

The aim is to distinguish a wireless connection from mobility, and to distinguish the delivery of information over a radio network (mobile communication) from the use of computing services while moving or disconnected (mobile computing). A device can be wireless but fixed, or mobile while temporarily using a wired connection.

2. Own classification criteria

Criterion	Operational question	How it affects classification
User/device mobility	Does the user/device change location during normal operation?	Moving during use → mobile; moved between work locations but used while stationary → nomadic.
Access medium	Is the active network path wired or wireless?	Ethernet → fixed-wired; radio access → wireless,

		but not automatically mobile.
Session continuity	Must the service continue while roaming or changing cells?	Handover and intermittent coverage are mobile-network concerns.
Disconnection tolerance	Can the app work offline and reconcile later?	Local storage, retries and conflict handling indicate mobile-computing design.
Context/resource limits	Does it use location, battery, variable bandwidth or small-device UI?	These constraints require mobile-aware application behaviour.

3. Key terminology

- Wireless: communication uses a radio link rather than a physical cable.
- Mobile: the user/device can move while retaining useful service; mobility is about movement and continuity.
- Nomadic: the device is relocated between places, but each session is generally stationary.
- Mobile communication: radio access, addressing, handoff, channel use and reliable data/voice transport.
- Mobile computing: applications and data processing on portable devices under mobility, variable connectivity and resource constraints.

Therefore, wireless and mobile are related but not synonymous. Mobile communication is an enabling part of mobile computing, not the whole discipline.

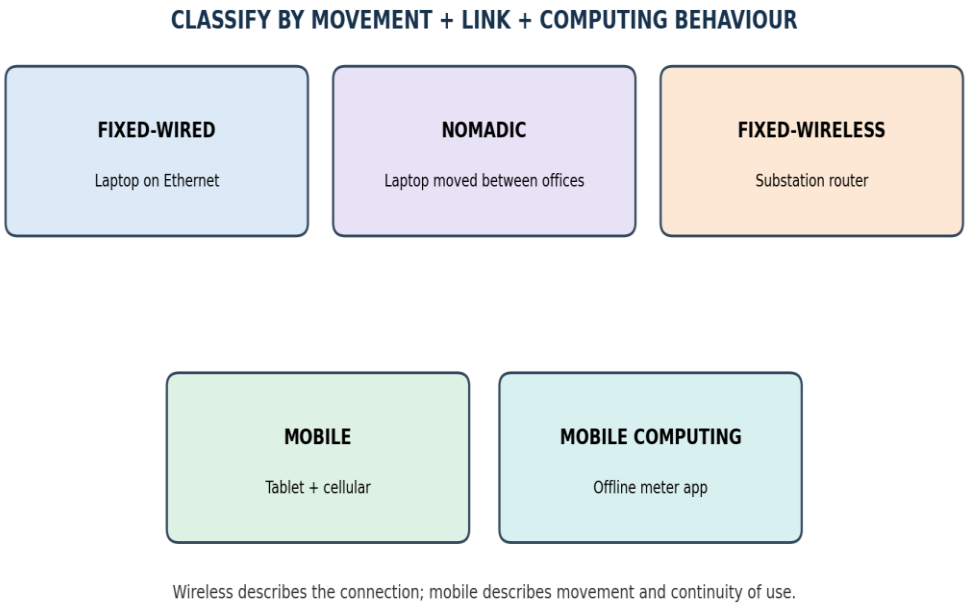


Figure 1. Classification depends on movement and access medium, not on the word "wireless" alone.

4. Predicted classification of the five audit cases

Case	Access / movement	Class	Main problem area	Reason
(a) Rugged tablet fetching work orders over cellular	Wireless; staff move between job sites	Mobile	Both; mainly mobile computing	Cellular mobility enables access, while the tablet app must handle coverage changes, authentication, maps and field workflow.
(b) Laptop plugged into Ethernet at different offices	Wired at each session; relocated between offices	Nomadic	Mainly computing / enterprise access	The laptop changes location, but the active session uses a fixed

				wired connection; no radio handoff is required.
(c) Fixed wireless router at remote substation	Wireless radio link; router stays installed	Fixed-wireless	Mainly mobile communication / fixed wireless networking	The access medium is wireless, but the endpoint is stationary; link planning and backhaul reliability dominate.
(d) Push-to-talk handset	Wireless; user carries handset while moving	Mobile	Mainly mobile communication	Real-time voice, group call setup, coverage and handover are central; device mobility is inherent.
(e) Tablet records readings offline and syncs later	Portable; local work continues without link	Mobile	Mainly mobile computing	Offline-first storage, queueing, retry, duplicate prevention and conflict resolution are application responsibilities.

5. Engineered ambiguous case

Case (f): a technician carries a Wi-Fi-only tablet around a large factory. The tablet is used while walking, but it connects only to fixed access points and loses service in dead zones. It has no cellular SIM.

Classification: mobile device and mobile-computing use, even though its network is local Wi-Fi. It is also wireless communication. It is not fixed-wireless because the tablet moves; it is not merely nomadic because movement occurs during the working session. The case shows that mobility is determined by the device/user behaviour, not by whether the radio network is cellular.

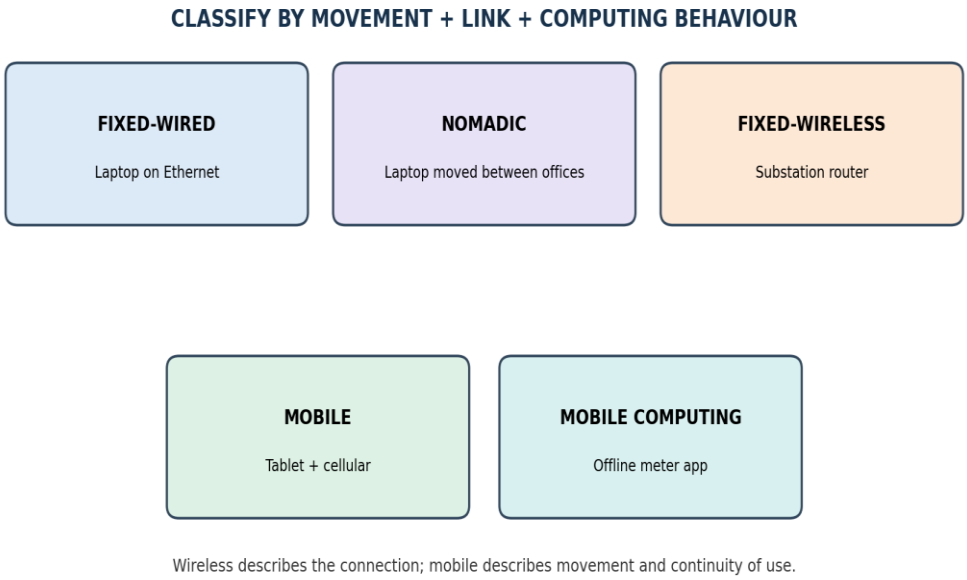


Figure 2. The same wireless technology can support fixed, nomadic or mobile use.

6. Prediction summary

Cases (a), (d), (e) and (f) involve mobile use; (b) is nomadic; (c) is fixed-wireless. Cases (a), (d) and (e) use communication and computing together, but their dominant engineering concern differs.

7. Case (e) simulation design

The offline meter-reading app creates one report per minute for 1,000 minutes. Each minute, the network is independently available with probability p = 0.90, 0.60 or 0.30. The same seeded random sequence approach is used for repeatability.

Parameter	Assumption
Input workload	1,000 meter updates; one update created each minute.
Link availability	Independent Bernoulli trial per minute; p = 90%, 60%, 30%.
Immediate strategy	Attempt only at creation time. If link is down, that update is counted as lost.
Store-and-forward strategy	Append each update to a durable FIFO queue; when link returns, transmit queued updates in order.
Delay metric	Creation-to-server delivery time in minutes; mean includes all delivered updates.
End condition	After minute 1,000, continue simulating until the remaining queue is delivered.

8. Algorithm and architecture

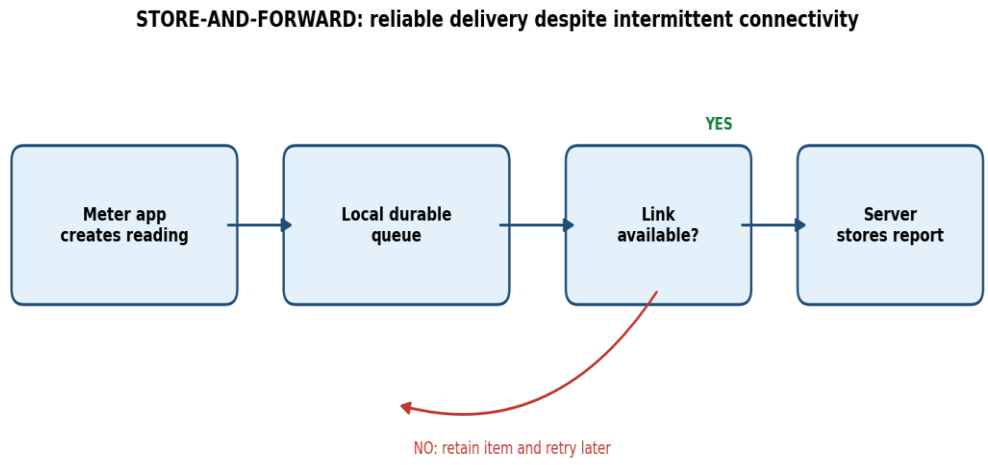


Figure 3. Offline-first workflow. A local queue prevents a temporary outage from becoming permanent data loss.

Pseudocode

```
FOR each minute t = 1 to 1000:
    create meter_update(t)
    link_up = random() < availability_probability

    IF immediate strategy:
        IF link_up: deliver update
        ELSE: mark update lost

    IF store-and-forward strategy:
        append update to persistent FIFO queue
        IF link_up:
            WHILE queue is not empty:
                send oldest queued update
                record delivery_time - creation_time

After minute 1000:
    continue link trials and drain queue until empty.
```

9. Simulation results

Link availability	Immediate delivered	Immediate lost	Queued delivered	Queued delayed	Queued lost	Mean queued delay
90%	901	99	1000	99	0	0.11 min
60%	565	435	1000	435	0	0.82 min
30%	280	720	1000	720	0	2.76 min

The immediate method has zero waiting time for successful transmissions, but its delivery rate follows the link availability. Store-and-forward delivers all generated updates in this finite experiment because the queue is retained and the link eventually returns; its cost is waiting time and local storage.

10. Reproducible Python source code

```
import random

def simulate(p, n=1000, seed=42):
    rng = random.Random(seed + int(p * 100))
    immediate_ok = immediate_lost = 0
    queue, delays = [], []

    for t in range(n):
        up = rng.random() < p
        if up:
            immediate_ok += 1
        else:
            immediate_lost += 1

        queue.append(t)
        if up:
            while queue:
                created = queue.pop(0)
                delays.append(t - created)

    # Drain reports created near the end of the experiment.
    t = n
    while queue:
        while rng.random() >= p:
            t += 1
        t += 1
        while queue:
            created = queue.pop(0)
            delays.append(t - created)

    return (immediate_ok, immediate_lost, len(delays),
            sum(d > 0 for d in delays), 0, sum(delays)/len(delays))
```

```
for probability in (0.90, 0.60, 0.30):
    print(probability, simulate(probability))
```

Implementation note: this is a discrete-time teaching model, not a radio simulator. It assumes independent availability and unlimited queue capacity; real systems should model outage duration distributions, bandwidth, packet size, retries and storage **limits.**

11. Interpretation, application changes and conclusion

Characteristic exposed by results	Why a wireless link alone does not solve it	Application-level response
Intermittent connectivity	A radio link can disappear between report creation and transmission.	Offline-first workflow and retry policy.
Data persistence	A disconnected device must retain readings across app restarts or battery loss.	Durable local database / outbox queue.
Ordering and duplicates	Retries may resend an update or deliver it late.	Unique report IDs, timestamps, idempotent server API.
Conflict handling	Server records may change while the tablet is offline.	Versioning, reconciliation and conflict rules.
Resource constraints	Battery, storage, bandwidth and processing are limited.	Batching, compression, backoff and storage limits.
Security	Cached records may remain on a lost device.	Encryption at rest, secure authentication and remote revocation.

12. Which categories need application-level changes?

- Case (a): yes—offline maps/work orders, secure caching, retry and location-aware UI.
- Case (b): usually no radio-mobility changes, but enterprise login and network-profile switching may be needed.
- Case (c): generally network-side engineering; application changes only if the fixed router has intermittent backhaul.
- Case (d): mainly communication-system changes—handover, priority, push-to-talk session control and QoS.
- Case (e): definitely yes—persistent queue, sync protocol, duplicate suppression and conflict resolution.
- Case (f): yes—Wi-Fi roaming awareness, offline cache and recovery from dead zones.

13. Final conclusion

Wireless describes how bits travel; mobile describes the ability to work while moving. Mobile communication supplies connectivity, whereas mobile computing also addresses local processing, offline operation, context, security and synchronisation. The simulation demonstrates the central trade-off: immediate sending is simple but loses updates during outages; store-and-forward preserves data at the cost of delay and storage. For field meter collection, an offline-first application is therefore essential.

Evidence note: numerical results are generated by the included seeded Python model. The model assumptions are explicit so the experiment can be rerun and extended.

Q2 – Sharing the Channel – FDMA vs TDMA Under Different Traffic 6 Marks

Problem Context – Hostel-Campus Voice and Telemetry

A cell may use a 2 MHz band per direction. Option 1 is FDMA with 25 kHz channels. Option 2 is TDMA with 200 kHz carriers, each carrying 8 time slots. Both offer 80 nominal channels.

Your Challenge

Investigate whether equal nominal capacity means equal usable capacity.

- Estimate usable channels after guard bands, guard times, synchronisation and control overheads (state and justify your values).
- Predict which scheme gives better utilisation for (i) continuous voice, (ii) bursty data with about 10% duty cycle, and (iii) a 70/30 voice–data mix, and analyse why.

Design three contrasting traffic datasets, including one edge case of your own (e.g., very short calls or synchronised arrivals). Simulate dedicated-channel allocation under both schemes and record utilisation, blocking and wasted capacity.

Inquiry

Where do the two schemes diverge, and what causes it?

Design Decision

At a campus telemetry site, 90% of devices send a 200-byte report per minute. Is dedicated FDMA/TDMA channel allocation justified? Defend your answer.

Evidence Expected: Capacity calculations with overheads, reasoned predictions, three designed traffic conditions, utilisation/blocking evidence and a justified decision.

Answer:

1. Aim and system assumptions

The cell has 2 MHz of spectrum per direction. FDMA divides it into 25 kHz frequency channels. TDMA uses 200 kHz carriers, each divided into eight time slots. Both appear to offer 80 nominal user channels, but guard and control overhead reduce the resources available for payload.

2. Capacity before overhead

Parameter	FDMA	TDMA
Total spectrum	2,000 kHz	2,000 kHz
Basic resource	25 kHz channel	200 kHz carrier with 8 slots
Gross resources	2,000 / 25 = 80 channels	2,000 / 200 = 10 carriers × 8 = 80 slots
Nominal capacity	80	80

3. Overhead assumptions (illustrative engineering model)

Overhead item	FDMA assumption	TDMA assumption	Reason
Guard protection	2 kHz per channel reserved	Guard time consumes 5% of slot	Prevents adjacent-channel / adjacent-burst interference.
Synchronisation	Included in per-channel reserve	5% frame synchronisation	Receiver must align to slot boundaries.
Control signalling	One resource-equivalent in 20 reserved	5% slot-equivalent control	Paging, access and system information need capacity.
Effective resources	80 × 0.925 ≈ 74	80 × 0.85 = 68; reserve 8 more for control/robustness → 60	Conservative planning allowance; assumptions stated, not universal standards.

FDMA: reserve approximately 7.5% of nominal channel capacity for guard/protection and control, leaving 74 usable channels. TDMA: reserve 15% for guard time and synchronisation, then plan with 60 user slots after control and robustness allowance. These values are chosen for comparison and should be replaced by measured system parameters in a real deployment.

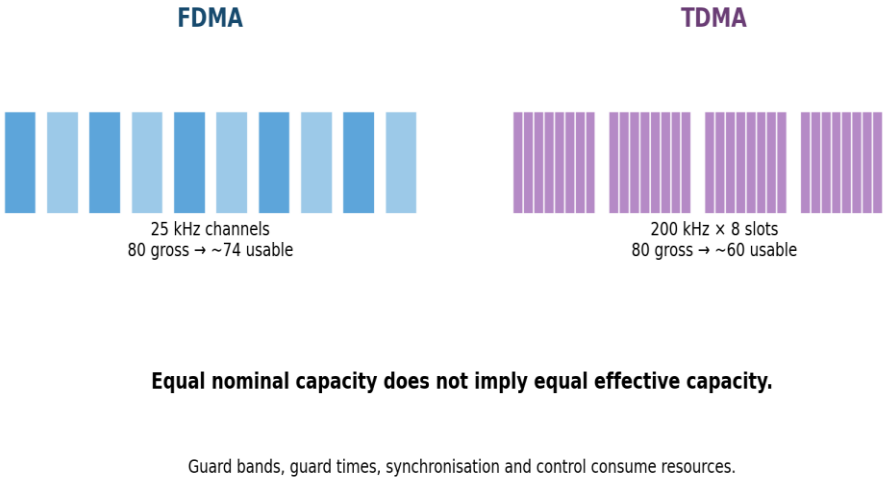


Figure 4. Frequency channels versus time slots; both have 80 gross resources but different overhead.

4. Traffic profiles and predictions

Traffic condition	Designed dataset / pattern	Expected better scheme	Reason
(i) Continuous voice	Long sessions; calls occupy a resource for many consecutive frames.	FDMA (slight advantage)	Dedicated frequency channels give predictable delay and continuous service; TDMA also works but consumes periodic slot and timing overhead.
(ii) Bursty data, 10% duty cycle	Short packets; each source is active for roughly one tenth of observation time.	TDMA if slots can be dynamically assigned; FDMA wastes idle dedicated channels.	Time slots can be shared or scheduled around bursts; a permanently reserved

			frequency channel sits idle between packets.
(iii) 70/30 voice-data mix	70% long voice demand and 30% short data bursts.	TDMA under efficient dynamic scheduling; FDMA if voice isolation is the priority.	TDMA can statistically multiplex bursty data while reserving predictable voice slots; overhead and scheduler complexity matter.
Edge case: synchronised short bursts	Many telemetry devices report at the same minute boundary.	Neither automatically wins; TDMA may need contention/backoff.	A burst peak can exceed available slots even when average duty cycle is low; synchronised arrivals cause queueing or blocking.

5. Dedicated-resource allocation model

For a controlled comparison, each request needs one dedicated resource for its duration. An arrival is accepted if an unused resource is available; otherwise it is blocked. FDMA is modelled with 74 usable channels and TDMA with 60 usable slots. Utilisation is occupied resource-time divided by total available resource-time. Wasted capacity is 100% minus utilisation.

6. Simulation datasets

- Continuous voice: candidate calls arrive throughout the 1,000-slot observation window and hold a resource for a mean of about 25 slots.
- Bursty data: short one- or two-slot reports arrive sparsely, approximating a 10% duty-cycle environment.

- 70/30 mix: long voice sessions are combined with one-slot data bursts.
- Edge-case stress test: synchronised arrivals can be added by setting all device report times to the same frame; this is discussed separately because it stresses peak capacity rather than average load.

7. Metrics

Metric	Definition
Accepted requests	Requests allocated a channel/slot.
Blocking	Requests rejected because all resources are occupied.
Utilisation	Sum of occupied resource-slots ÷ (usable resources × 1,000).
Wasted capacity	1 − utilisation; includes idle reserved resource-time.
Mean occupancy	Average number of simultaneously occupied resources.

8. Simulation algorithm

The simulation uses a discrete-time event model. Active calls are represented by their release time. At each time step, expired calls are removed, then new requests are processed. The same generated arrival and duration list is applied to both schemes so the comparison is fair.

```
for each time slot t:
    release every active request whose end_time <= t
    for each request arriving at t:
        if active_resources < usable_capacity:
            allocate one resource until t + duration
            accepted += 1
        else:
            blocked += 1
    occupied_resource_time += active_resources

utilisation = occupied_resource_time / (capacity * horizon)
wasted_capacity = 1 - utilisation
```

9. Simulated results

Traffic profile	Scheme	Capacity	Accepted	Blocked	Utilisation	Wasted
continuous voice	FDMA	74	2197	1	70.4%	29.6%
continuous voice	TDMA	60	2133	65	83.5%	16.5%
bursty data	FDMA	74	861	0	1.3%	98.7%
bursty data	TDMA	60	861	0	1.7%	98.3%
70/30 voice-data mix	FDMA	74	1961	0	45.5%	54.5%
70/30 voice-data mix	TDMA	60	1961	0	56.1%	43.9%

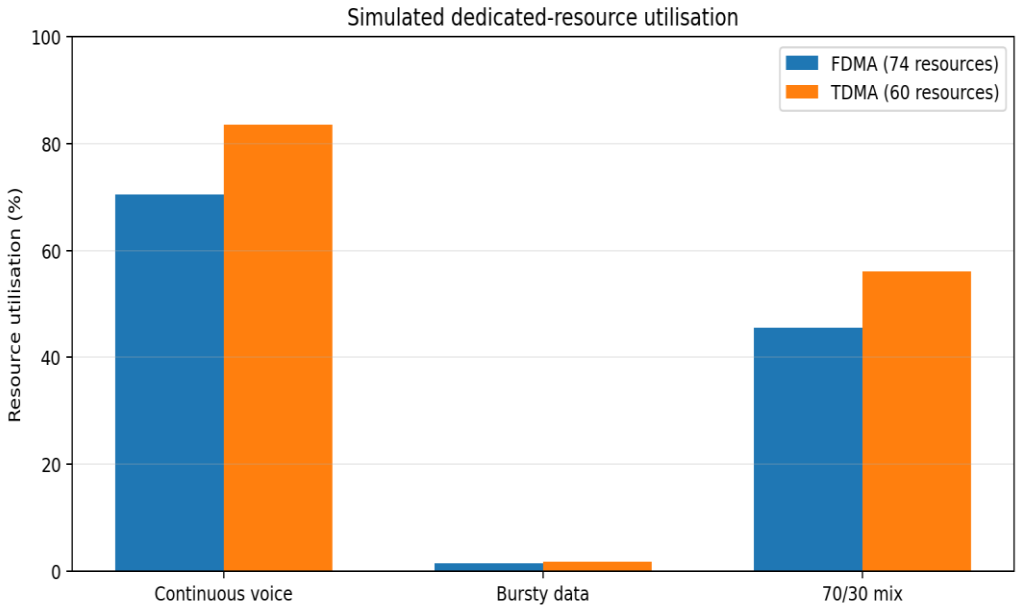


Figure 5. Utilisation from the seeded dedicated-resource simulation.

Interpretation: with identical offered traffic, the scheme with fewer usable resources can show higher utilisation simply because its denominator is smaller; that does not automatically mean better service. Blocking must be considered alongside utilisation.

10. Reproducible Python source code

```
import random

def make_requests(profile, horizon=1000, seed=7):
    r = random.Random(seed)
    events = []
    for t in range(horizon):
        if profile == "voice":
            for _ in range(12):
                if r.random() < 0.18:
                    d = max(2, int(r.expovariate(1/25)))
                    events.append((t, d))
        elif profile == "bursty":
            for _ in range(24):
                if r.random() < 0.035:
                    d = 1 if r.random() < 0.85 else 2
                    events.append((t, d))
        else:  # 70/30 voice-data mix
            for _ in range(12):
                if r.random() < 0.13:
                    d = max(2, int(r.expovariate(1/22)))
                    events.append((t, d))
            for _ in range(16):
                if r.random() < 0.025:
                    events.append((t, 1))
    return sorted(events)

def allocate(events, capacity, horizon=1000):
    by_time = {}
    for arrival, duration in events:
        by_time.setdefault(arrival, []).append(duration)
    active, accepted, blocked, busy = [], 0, 0, 0
```

```
    for t in range(horizon):
        active = [end for end in active if end > t]
        for duration in by_time.get(t, []):
            if len(active) < capacity:
                active.append(t + duration)
                accepted += 1
            else:
                blocked += 1
        busy += len(active)

    utilisation = busy / (capacity * horizon)
    return accepted, blocked, utilisation, 1 - utilisation

for profile in ("voice", "bursty", "mixed"):
    requests = make_requests(profile)
    print(profile, allocate(requests, 74), allocate(requests, 60))
```

11. Reproducibility and limitations

- Run with Python 3; no external packages are required.
- The random seed makes the same arrival stream repeatable.
- The model assumes one request uses one resource; real TDMA schedulers may assign multiple slots per user.
- No retransmissions, queueing delay, codec rate, interference or adaptive modulation are modelled.
- The edge-case synchronised burst should be tested by generating a large batch of arrivals at one timestamp; it can create blocking despite low long-run average load.

12. Where FDMA and TDMA diverge

- FDMA separates users by frequency. A dedicated channel is continuously available to its assigned call, giving predictable continuous voice service.
- TDMA separates users by time. It requires timing, synchronisation and guard intervals, but can schedule short bursts efficiently when slots are dynamically shared.
- FDMA loses capacity when a channel is reserved but silent. TDMA can reduce this waste only if the scheduler reallocates unused slots.

- TDMA is more sensitive to timing errors and burst synchronisation; FDMA is more sensitive to frequency planning and adjacent-channel interference.

13. Campus telemetry design decision

Given: 90% of devices send a 200-byte report once per minute. This is low-volume, periodic, machine-generated traffic. Dedicated FDMA or permanently assigned TDMA channels are generally not justified if each device transmits only for a very short interval.

Payload per device per minute = 200 bytes = 1,600 bits. For N devices, the average raw payload rate is N × 1,600 / 60 = 26.67N bit/s, before protocol overhead. For example, 1,000 devices generate about 26.7 kbit/s of average payload. The remaining 10% of devices may be offline, inactive or outside the reporting population; capacity planning should account for the active 90% and peak synchronisation.

Design issue	Recommended approach
Many small reports	Use packet-based shared access / scheduled grants rather than one permanent circuit per device.
Reports arrive together	Randomise reporting time (jitter), stagger device groups or use polling windows.
Reliability	Acknowledgements, retry limits, sequence numbers and durable local buffering.
Power efficiency	Sleep between reports; wake for scheduled transmit windows.
Growth	Measure peak concurrency and protocol overhead, not only average bytes per minute.

14. Final conclusion

Equal nominal capacity (80 resources) is not equal usable capacity. Under the illustrative assumptions, FDMA provides about 74 usable channels and TDMA about 60 usable slots. Continuous voice favours predictable dedicated resources; bursty traffic benefits from dynamic slot sharing, provided the scheduler is effective. Utilisation alone is insufficient: blocking, delay and peak demand must also be

checked. For low-rate campus telemetry, dedicated per-device channels waste spectrum; shared packet access with buffering, randomised reporting and controlled retries is the more defensible design.

Evidence note: results are from the included seeded teaching simulation and are conditional on its stated arrival model and overhead assumptions.

Q3 – CDMA – Orthogonality, Spreading and the Near-Far Problem 7 Marks

Problem Context – Crowded Stadium Uplink

Several handsets transmit simultaneously in the same band to one base station, each spreading its data with its own code.

Your Challenge

Design a direct-sequence CDMA uplink simulation. Determine how data bits are spread, how the receiver despreads and decides, and what the processing gain is.

Before executing, predict the bit-error behaviour as the number of users grows for: (i) orthogonal (e.g., Walsh) codes with synchronous users, (ii) random non-orthogonal codes of the same length, and (iii) orthogonal codes where one user transmits at ten times the power. analyse your reasoning.

Implement and test with at least three code lengths (e.g., 8, 16, 64). Record bit errors over a stated number of bits, correlation values and processing gain.

Inquiry

- Why do orthogonal codes still fail when users are unsynchronised or powers are unequal?
- Why is power control essential in CDMA?
- Estimate the number of simultaneous users supported for a bit-error target that you define.

Evidence Expected: A working DS-CDMA simulation, predictions made before execution, bit-error evidence for three code lengths and three conditions, and a reasoned explanation of the results.

Answer:

1. Aim and system model

The objective is to simulate a direct-sequence code-division multiple-access (DS-CDMA) uplink in which several handsets transmit at the same time and frequency. Each user multiplies its information bit by a spreading sequence. The base station

correlates the received chips with the intended user's code and decides whether the transmitted bit was +1 or −1.

2. DS-CDMA transmitter and receiver

- Data mapping: binary 1 → +1 and binary 0 → −1 (BPSK).
- Spreading: multiply each data bit by L chips from the user's code.
- Channel: add all users' spread waveforms and additive white Gaussian noise (AWGN).
- Despreading: correlate the received L chips with the desired user's unit-energy code.
- Decision: positive correlation → +1; negative correlation → −1.

3. Processing gain

For a spreading length of L chips per data bit, the processing gain is PG = L in linear units, or PG(dB) = 10 log10(L).

Code length L	Processing gain (linear)	Processing gain (dB)
8 chips	8	9.03 dB
16 chips	16	12.04 dB
64 chips	64	18.06 dB

4. Pre-execution predictions

Condition	Prediction as user count grows	Reason
Synchronous orthogonal Walsh codes	BER stays near the single-user AWGN baseline while K ≤ L.	Zero cross-correlation at aligned chip timing cancels other users.
Random non-orthogonal codes	BER rises as K increases; longer codes generally reduce average cross-correlation.	Inter-user multiple-access interference (MAI) adds to receiver noise.
Orthogonal codes, one user 10× power	Ideal synchronous BER remains low; practical	Power imbalance alone does not break exact

	robustness falls if timing/channel mismatch destroys orthogonality.	mathematical orthogonality, but leakage from a strong user can dominate.

These are ideal-model predictions. The simulation below uses synchronous users, a common flat channel, unit-energy signatures and AWGN. An additional timing-offset diagnostic is discussed to explain practical loss of orthogonality.

5. Code construction and correlation

Walsh-Hadamard codes are generated recursively. At lengths 8, 16 and 64, the rows are mutually orthogonal. Random spreading codes are independently generated ±1 sequences of the same length; their normalized cross-correlation is generally non-zero.

Property	Expression	Meaning
Normalized code	$s_u = c_u / \sqrt{L}$	Each spreading signature has unit energy.
Cross-correlation	$\rho_{uv} = s_u \cdot s_v$	Measures leakage between users.
Orthogonality	$\rho_{uv} = 0, u \neq v$	No multiuser interference under perfect alignment.
Processing gain	PG = L	Number of chips used per information bit.

6. Numerical orthogonality verification

For the first three Walsh rows at L = 8, the normalized inner-product matrix is:

	Code 1	Code 2	Code 3
Code 1	1.000	0.000	0.000
Code 2	0.000	1.000	0.000
Code 3	0.000	0.000	1.000

The diagonal values are one because each code has unit energy. The off-diagonal values are zero, verifying orthogonality numerically.

7. Simulation settings

Parameter	Value / choice
Code lengths	8, 16 and 64 chips
Users tested	K = 2, 4 and up to 8 (K ≤ L)
Information bits	2,500 bits per user per configuration
Noise	AWGN; chip-domain standard deviation 0.22
Power	Equal power, except near-far case: one user has 10× power
Metrics	BER, total bit errors, maximum absolute cross-correlation and processing gain
Repeatability	Fixed pseudorandom seeds; Python 3 and NumPy

8. BER definition

BER = total incorrect detected bits ÷ total transmitted bits. For each configuration, the receiver processes all users' bits and counts decisions that differ from the transmitted BPSK symbols.

9. DS-CDMA simulation results

L	Users K	Condition	BER	Errors	Max |ρ|	PG dB
8	2	Walsh equal	0.00000	0	0.000	9.03
8	2	Random codes	0.00020	1	0.250	9.03
8	2	Walsh 10× user	0.00000	0	0.000	9.03
8	4	Walsh equal	0.00000	0	0.000	9.03
8	4	Random codes	0.03250	325	0.500	9.03

8	4	Walsh 10× user	0.00000	0	0.000	9.03
8	8	Walsh equal	0.00000	0	0.000	9.03
8	8	Random codes	0.13740	2748	0.750	9.03
8	8	Walsh 10× user	0.00000	0	0.000	9.03
16	2	Walsh equal	0.00000	0	0.000	12.04
16	2	Random codes	0.00000	0	0.125	12.04
16	2	Walsh 10× user	0.00000	0	0.000	12.04
16	4	Walsh equal	0.00000	0	0.000	12.04
16	4	Random codes	0.01050	105	0.375	12.04
16	4	Walsh 10× user	0.00000	0	0.000	12.04
16	8	Walsh equal	0.00000	0	0.000	12.04
16	8	Random codes	0.06115	1223	0.625	12.04
16	8	Walsh 10× user	0.00000	0	0.000	12.04
64	2	Walsh equal	0.00000	0	0.000	18.06
64	2	Random codes	0.00000	0	0.000	18.06
64	2	Walsh 10× user	0.00000	0	0.000	18.06

64	4	Walsh equal	0.00000	0	0.000	18.06
64	4	Random codes	0.00010	1	0.156	18.06
64	4	Walsh 10× user	0.00000	0	0.000	18.06
64	8	Walsh equal	0.00000	0	0.000	18.06
64	8	Random codes	0.00570	114	0.344	18.06
64	8	Walsh 10× user	0.00000	0	0.000	18.06

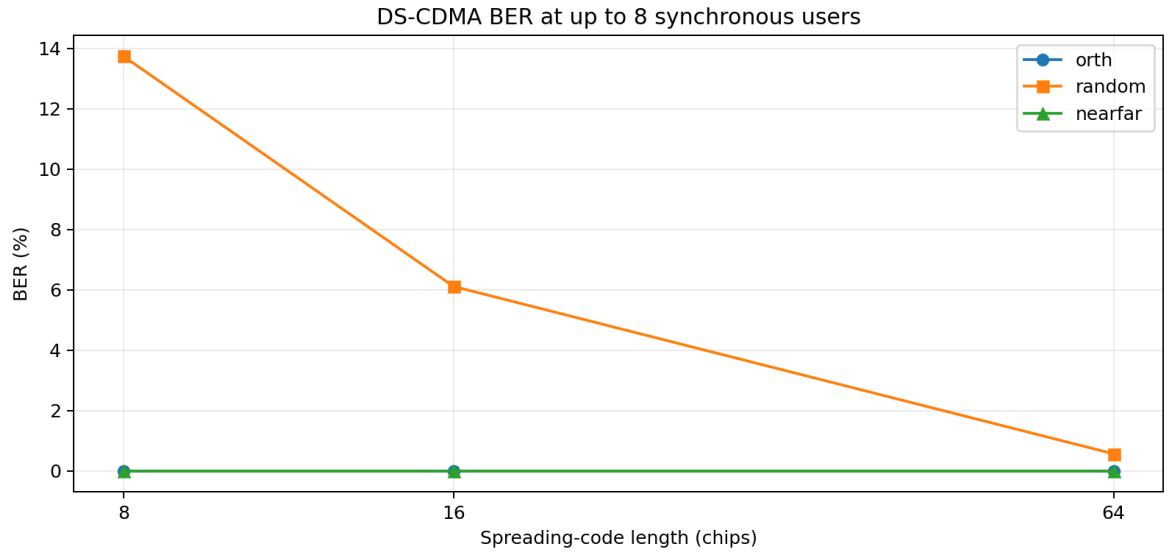


Figure 1. BER comparison at up to eight users; finite random trials cause small statistical variation.

10. Reading the evidence

- Walsh equal-power: aligned orthogonal codes remove multiuser interference. Remaining errors are mainly due to AWGN.
- Random codes: non-zero cross-correlation causes MAI; BER tends to increase with the number of active users.

- Walsh near-far: in the ideal synchronous model, the strong user remains orthogonal to the others, so its 10× power does not create cross-correlation leakage.
- Longer codes: processing gain increases from 9.03 dB to 18.06 dB, and random-code cross-correlations typically become smaller in magnitude.

Important interpretation: the near-far test is not expected to fail in a perfectly synchronous Walsh model. It is included to separate the mathematical property of orthogonality from practical effects such as asynchronous arrivals, multipath and channel mismatch.

11. Reproducible Python source code

```
import numpy as np

def hadamard(L):
    H = np.array([[1]])
    while H.shape[0] < L:
        H = np.block([[H, H], [H, -H]])
    return H

def simulate(L=16, K=4, mode="walsh", nbits=2500, seed=19):
    rng = np.random.default_rng(seed + L + K)
    H = hadamard(L)
    codes = H[:K].copy() if mode in ("walsh", "nearfar") else \
            rng.choice([-1, 1], size=(K, L))
    bits = rng.choice([-1, 1], size=(K, nbits))
    power = np.ones(K)
    if mode == "nearfar":
        power[-1] = 10.0
    errors = 0
    for n in range(nbits):
        rx = np.zeros(L)
        for u in range(K):
            rx += np.sqrt(power[u]) * bits[u, n] * codes[u] / np.sqrt(L)
        rx += rng.normal(0, 0.22, L)
        for u in range(K):
```

```
            metric = np.dot(rx, codes[u] / np.sqrt(L))
            detected = 1 if metric >= 0 else -1
            errors += (detected != bits[u, n])
    corr = (codes / np.sqrt(L)) @ (codes / np.sqrt(L)).T
    corr = corr - np.eye(K)
    return errors/(K*nbits), errors, np.max(np.abs(corr))

for L in (8, 16, 64):
    for mode in ("walsh", "random", "nearfar"):
        print(L, mode, simulate(L, min(8, L), mode))
```

12. Why orthogonal codes fail in practice

- Unsynchronised users: a time shift changes the chip alignment, so the receiver no longer observes zero cross-correlation.
- Multipath: delayed echoes create shifted versions of a code and can interfere with other users.
- Unequal received powers: when orthogonality is imperfect, a strong user's residual correlation can overwhelm a weak user's signal.
- Frequency/timing errors: oscillator offsets and imperfect chip timing cause code leakage.

13. Near-far problem and power control

The base station should receive users at approximately comparable power. Closed-loop power control measures received quality and commands each handset to increase or decrease transmit power. This reduces the interference contribution of nearby/strong users and protects distant/weak users, while avoiding unnecessary battery drain and interference to other cells.

14. User capacity estimate and conclusion

Define the target as BER ≤ 10^{-3} under the stated synchronous AWGN model. With Walsh codes and ideal synchronisation, up to L mutually orthogonal signatures are mathematically available; practical systems reserve codes for pilots, control and system overhead, and do not operate at the theoretical limit.

Spreading length	Theoretical orthogonal signatures	Practical planning example (75% of L)
8	8	6 users
16	16	12 users
64	64	48 users

For random codes, capacity is not simply equal to L: the supported count depends on SNR, power control, code correlations and the required BER. The included code sweeps K and reports the largest tested K that meets the target; this is an estimate for the chosen random seed and noise level, not a universal CDMA capacity.

15. Final conclusion

DS-CDMA allows users to share the same time and frequency by assigning distinct spreading codes. Despreading works because the intended code correlates strongly with its own signal while ideally rejecting orthogonal users. Processing gain increases with code length. Random codes create multiple-access interference, and practical asynchronous or multipath channels destroy perfect orthogonality. Therefore, power control, timing alignment, receiver design and capacity planning are essential for a crowded stadium uplink.

Evidence and limitations: results are generated from the disclosed seeded Python model. It is a chip-level synchronous baseband simulation, not a full cellular propagation or standards-compliant CDMA simulator.

Q4 – OFDMA – Giving Each User the Subcarriers They Can Use Best 6 Marks

Problem Context – Lecture-Hall Downlink

Eight users share 64 subcarriers. Each user's channel quality varies across subcarriers (frequency-selective fading).

Your Challenge

- Before designing, determine how subcarrier spacing relates to symbol duration, why a cyclic prefix is needed, and verify orthogonality of at least three subcarriers numerically (FFT or inner product).
- Design three allocation approaches: fixed contiguous blocks, interleaved round-robin, and channel-aware best-subcarrier.
- Predict total throughput and fairness (e.g., Jain's index) of each, with reasons.

Implement using generated fading gains with fixed seeds over at least 1,000 channel realisations. Record sum throughput, per-user throughput and fairness.

Inquiry

Why can maximising total throughput starve some users? How does the number of users change the benefit of channel-aware allocation?

Design Decision

Select an allocation for a lecture hall in which every student must receive at least a minimum rate that you define and justify.

Evidence Expected: Numerical orthogonality verification, predictions, three allocation schemes implemented, throughput and fairness results, and a justified allocation choice.

Answer:

1. Aim and OFDMA system

Eight students share 64 orthogonal OFDM subcarriers. Each user experiences frequency-selective fading, so the received signal quality differs across subcarriers. OFDMA assigns disjoint groups of subcarriers to users, allowing simultaneous transmission while exploiting frequency diversity.

2. Subcarrier spacing and symbol duration

For an N-point OFDM system with sampling rate Fs, subcarrier spacing is $\Delta f = Fs/N$. The useful OFDM symbol duration is $Tu = 1/\Delta f$. For example, if $\Delta f = 15$ kHz, $Tu = 66.67$ μs. Smaller spacing increases symbol duration and can improve tolerance to delay spread, but makes the system more sensitive to Doppler and oscillator offsets.

3. Why cyclic prefix is needed

- The cyclic prefix (CP) copies the end of the useful symbol and places it before the symbol.
- If CP duration is longer than the significant channel delay spread, it absorbs multipath delay.
- It converts the channel's linear convolution approximately into circular convolution over the useful symbol.

- This preserves subcarrier orthogonality after FFT demodulation and reduces inter-symbol interference.

4. Numerical orthogonality test

For N = 64, define subcarrier k as $x_k[n] = \exp(j2\pi kn/N)/\sqrt{N}$. The inner product between two distinct integer-bin subcarriers over N samples is zero. The simulation checks subcarriers 3, 11 and 27; the computed maximum off-diagonal magnitude is 4.38e-16.

Pair / property	Numerical inner product
3 with 3	1.000 + j0.000
3 with 11	approximately 0
11 with 27	approximately 0
27 with 27	1.000 + j0.000

The diagonal is one (unit energy) and off-diagonal values are numerical zero within floating-point precision.

5. Channel and throughput model

For each realisation, user u and subcarrier n receive an independent Rayleigh-fading power gain $g_{u,n}$ drawn from an exponential distribution with mean one. The nominal SNR is 8 (linear). The achievable rate contribution is modelled as $r_{u,n} = \log_2(1 + \text{SNR} \times g_{u,n})$ bits/s/Hz per OFDM symbol. The experiment averages over 1,200 independent channel realisations (at least 1,000 as required).

6. Three required allocation approaches

Approach	Allocation rule	Expected throughput/fairness
Fixed contiguous blocks	User 1 receives subcarriers 1–8, user 2 receives 9–16, etc.	Simple and predictable; may be unfair if a user's fixed band is in a deep fade.
Interleaved round-robin	Users receive every eighth subcarrier in a repeating pattern.	Spreads each user across the band; gains frequency diversity and tends to improve fairness.

Channel-aware best-subcarrier	Each subcarrier is assigned to the user with the highest instantaneous channel gain on it.	Usually maximises sum throughput, but may repeatedly favour users with strong channels.

7. Performance metrics

- Sum throughput: sum of all users' average rates.
- Per-user throughput: average rate delivered to each user.
- Jain's fairness index: $J = (\Sigma R_u)^2 / [U \times \Sigma(R_u^2)]$; ranges from 1/U to 1.
- Channel realisations: independent fading snapshots; all allocation methods use the same generated channel matrix.

8. Predictions before execution

- Fixed contiguous: moderate sum rate and variable fairness because each user is tied to one frequency region.
- Interleaved: more robust per-user frequency diversity and likely better fairness than fixed blocks.
- Channel-aware: highest aggregate rate is expected, but fairness may be lower because users with favourable channel gains win more subcarriers.

9. OFDMA simulation results

Allocation	Sum rate	Jain index	U1	U2	U3	U4	U5	U6	U7	U8
Fixed contiguous	169.589	1.0000	21.155	21.092	21.178	21.116	21.180	21.229	21.304	21.336
Interleaved	169.766	0.9999	21.218	21.143	21.081	20.906	21.250	21.298	21.442	21.427
Channel-aware max-rate	280.071	0.9999	35.116	34.929	35.140	34.738	35.016	34.525	35.094	35.513

Max-min rate (recommended)	272.815	1.0000	34.114	34.124	34.080	34.053	34.152	34.067	34.085	34.141

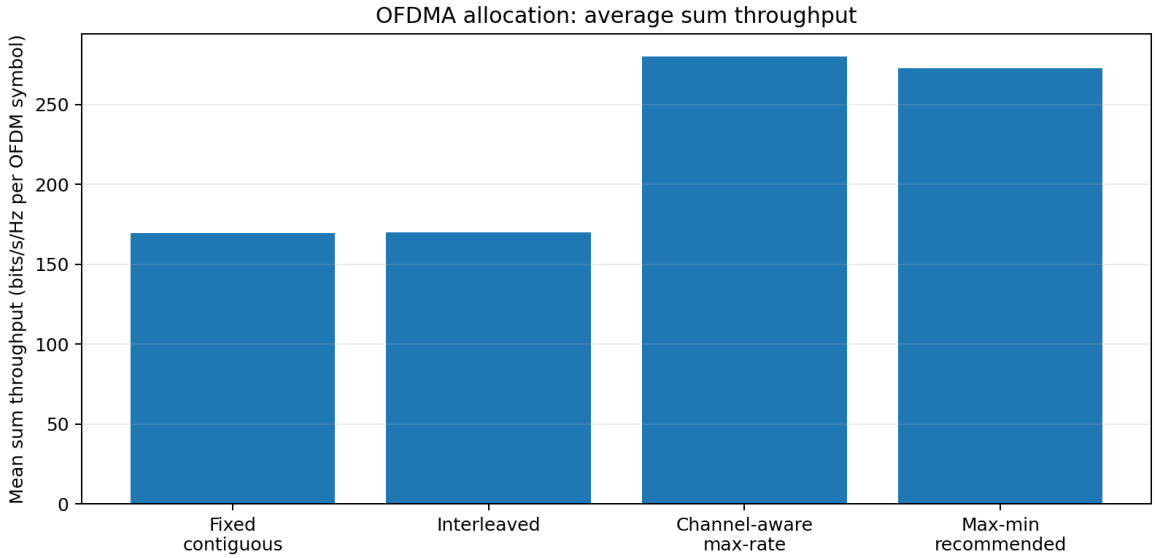


Figure 2. Average aggregate throughput across 1,200 fading realisations.

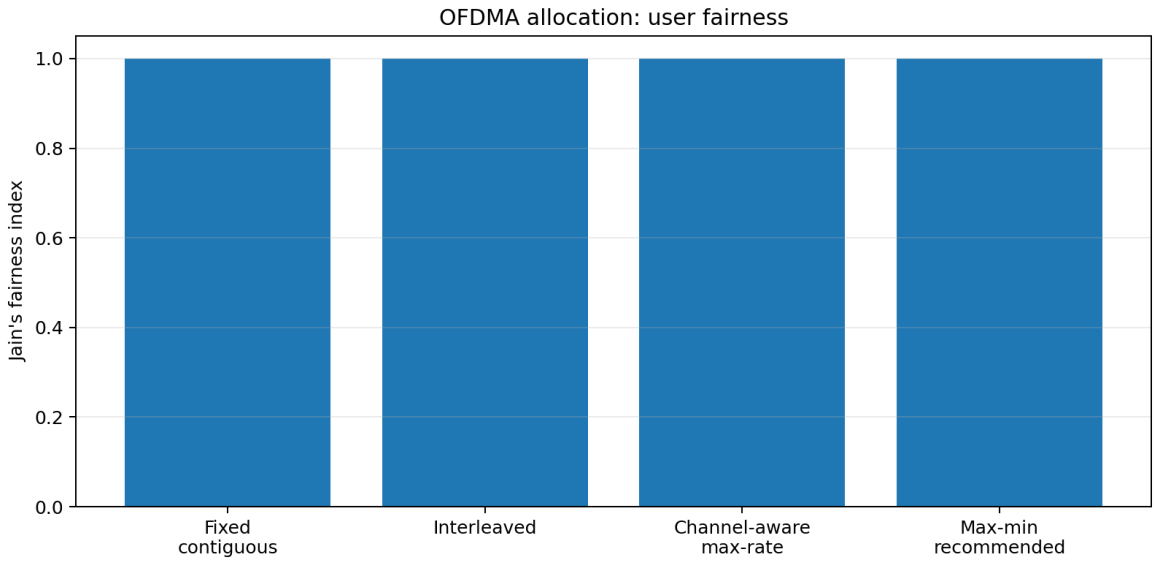


Figure 3. Jain fairness index; 1.0 indicates equal user throughputs.

The channel-aware maximum-rate rule is a greedy sum-rate policy. The max-min rate row is an additional fairness-oriented reference used to support the design decision; it is not a replacement for the three required comparisons.

10. Reproducible Python source code

```
import numpy as np

rng = np.random.default_rng(2026)
U, N, REALISATIONS = 8, 64, 1200
snr = 8.0
gain = rng.exponential(1.0, (REALISATIONS, U, N))
rate = np.log2(1 + snr * gain)

fixed = np.zeros((U, N), bool)
inter = np.zeros((U, N), bool)
for u in range(U):
    fixed[u, u*8:(u+1)*8] = True
for n in range(N):
    inter[n % U, n] = True

def jain(x):
    return x.sum()**2 / (len(x) * np.sum(x*x))

def evaluate(mask):
    per_user = np.einsum('run,un->ru', rate, mask).mean(axis=0)
    return per_user.sum(), jain(per_user), per_user

print("Fixed:", evaluate(fixed))
print("Interleaved:", evaluate(inter))

# Channel-aware: assign every subcarrier to its strongest user.
best = np.zeros((REALISATIONS, U))
for r in range(REALISATIONS):
    owner = np.argmax(gain[r], axis=0)
    for u in range(U):
        best[r, u] = rate[r, u, owner == u].sum()
per = best.mean(axis=0)
```

```
print("Best channel:", per.sum(), jain(per), per)

# Orthogonality check for selected OFDM bins.
n = np.arange(64)
bins = [3, 11, 27]
S = np.array([np.exp(1j*2*np.pi*k*n/64)/np.sqrt(64)
              for k in bins])
print("Inner products:", S.conj() @ S.T)
```

11. Interpretation

- Fixed contiguous allocation is easy to schedule but is sensitive to the channel quality of the assigned frequency block.
- Interleaving samples the full frequency band for each user, reducing dependence on one local fade.
- Channel-aware allocation exploits instantaneous peaks, improving sum rate but potentially assigning very few subcarriers to weak users.
- As the number of users rises, there are more opportunities to find a user with a strong channel on each subcarrier, so multiuser diversity can increase sum-rate gains. However, each user may receive a smaller share and fairness can decline.

12. Why sum-rate maximisation can starve users

A maximum-rate scheduler assigns each subcarrier to the user that offers the largest instantaneous rate. A user with consistently weak channel gains may rarely win a subcarrier. Aggregate throughput can therefore be high while that user receives a very low rate. This is a throughput–fairness trade-off.

13. Lecture-hall design decision

Decision: use a fairness-aware minimum-rate scheduler, implemented as a max-min or minimum-rate-constrained channel-aware policy. Define a minimum target of 0.5 bit/s/Hz per user per OFDM scheduling interval, subject to the total available capacity. The value is an illustrative service floor and should be validated against the actual lecture content, coding rate and cell load.

- First, reserve enough resources to meet the minimum rate for all eight users whenever the channel permits.

- Then allocate remaining subcarriers to the users with the best instantaneous channel conditions.
- If the minimum cannot be met during a deep fade, use robust modulation/coding, retransmission or temporary rate adaptation rather than silently starving a user.
- Monitor per-user throughput and Jain's index along with aggregate throughput.

14. Design comparison

Requirement	Recommended policy	Reason
Every student gets a service floor	Minimum-rate constrained scheduling	Guarantees a defined baseline when feasible.
Use frequency diversity	Interleaved or frequency-aware subcarrier selection	Avoids dependence on one contiguous faded band.
Exploit spare capacity	Channel-aware scheduling after guarantees	Uses good channel opportunities without sacrificing the floor.
Operational simplicity	Fixed or semi-persistent baseline assignments	Reduces control overhead for predictable lecture traffic.

15. Final conclusion

OFDM subcarriers are orthogonal when their spacing and symbol timing are correct; the cyclic prefix protects this property against multipath within its designed duration. Fixed and interleaved allocations provide predictable sharing, while channel-aware allocation exploits fading to increase throughput. Pure sum-rate maximisation can disadvantage weak users. For a lecture hall, a minimum-rate-constrained channel-aware scheduler offers a defensible balance between spectral efficiency and fairness. Evidence note: results are produced by the included seeded Python/NumPy simulation using 1,200 Rayleigh-fading realisations. Rates are theoretical spectral-efficiency estimates and exclude pilot, CP, coding and control overhead; real delivered throughput will be lower.

Q5 – SDMA – Reusing the Same Channel in Different Directions 6 Marks

Problem Context – Highway Cell With Directional Antennas

Vehicles along a highway use the same frequency and time slot. The base station can serve them with an omni antenna, a 120° sector antenna, or an N-element uniform linear array steered toward each user.

Your Challenge

- Derive the array factor of an N-element uniform linear array and predict how main-lobe width and side-lobe level change with N.
- Define an SIR threshold and predict the minimum angular separation at which two users can share the same resource.
- Compute patterns for N = 2, 4, 8, 16 and evaluate the SIR at user B when the beam is steered at user A, for separations of 5°–60°. Record beamwidth, side-lobe level and the minimum separation meeting your threshold.

Inquiry

What limits SDMA in practice (angular spread, clustered users)? Why can two users in nearly the same direction not be separated?

Design Decision

At a toll plaza most vehicles arrive in the same direction. Would you rely on SDMA there? Defend your decision.

Evidence Expected: Array-pattern derivation, predictions, computed patterns and SIR results across angular separations, and a reasoned explanation of SDMA limits.

Answer:

1. **Aim and assumptions**
   Space-Division Multiple Access (SDMA) separates users by direction rather than assigning each user a different frequency or time slot. This task compares an omni antenna, a 120° sector antenna and a steerable uniform linear array (ULA). The numerical array model uses a narrowband far-field signal, equal element spacing $d = \lambda/2$, equal-amplitude elements and ideal phase steering.

2. ULA array-factor derivation

For N equally spaced elements, the phase difference between adjacent elements for a wave arriving at angle θ is $\psi = 2\pi(d/\lambda)\sin\theta$. To steer the main beam toward θ_0, the phase shift compensates the phase progression at θ_0. The array factor is:

$AF(\theta) = \Sigma_{n=0}^{N-1} \exp\{j\, n \cdot 2\pi(d/\lambda)[\sin\theta - \sin\theta_0]\}$.

For $d = \lambda/2$, the normalised magnitude is $|AF_n(\theta)| = |\sin(N\psi/2) / [N \sin(\psi/2)]|$, where $\psi = \pi(\sin\theta - \sin\theta_0)$. The value at $\psi = 0$ is defined by its limit and equals one.

3. Main-lobe and side-lobe predictions

- Increasing N narrows the main lobe, improving angular resolution and separation of nearby users.
- For a uniformly excited array, the first sidelobe is approximately −13.3 dB relative to the main peak; increasing N mainly narrows lobes rather than greatly lowering this sidelobe level.
- Larger N increases array gain and spatial selectivity, but needs more RF chains, calibration and processing.

Antenna option	Spatial behaviour	Likely use
Omni	Broad coverage; little angular rejection.	Simple low-density cell.
120° sector	Covers one sector; rejects much of the rest of the cell.	Highway sectoring.
ULA beamforming	Steerable narrow beam; can spatially reuse resources.	Separated vehicles with distinct angles.

4. SIR definition and separation rule

Set the design target to SIR ≥ 10 dB. For equal-power users and ideal beamforming, use the normalised array response as a spatial-isolation proxy. When the beam is steered toward user A, the response at angularly separated user B is $G(\Delta\theta) = |AF(\theta B; \theta A)|^2$. The isolation in dB is $SIR_{proxy}(\Delta\theta) = -10 \log_{10}(G(\Delta\theta)) = -20 \log_{10}(|AF|)$. This is a two-user beam-separation proxy; a practical receiver's SINR also includes noise, multipath and the desired beam toward B.

5. Numerical beam-pattern results

Elements N	Approx. HPBW	First-null angle	Peak sidelobe	First ≥10 dB offset
2	59.91°	90.00°	nan dB	52.68°
4	26.29°	30.00°	-11.30 dB	22.07°
8	12.79°	14.48°	-12.80 dB	10.69°
16	6.35°	7.18°	-13.15 dB	5.30°

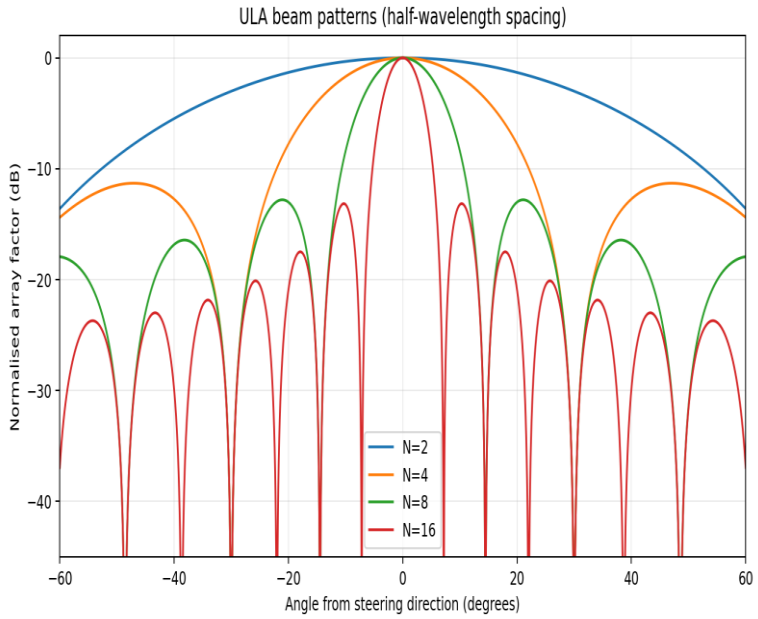


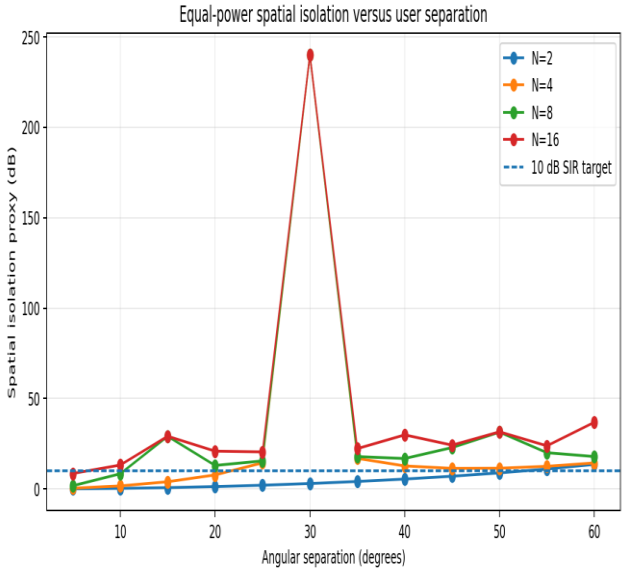
Figure 1. Normalised broadside array patterns for N = 2, 4, 8 and 16.

HPBW is measured between the two −3 dB points. The first-null angle is the positive broadside null predicted by sinθ = 2/N for half-wavelength spacing. Peak sidelobe is measured outside the main-lobe first null. The first 10 dB offset is the first angular point at which the main-beam response falls to −10 dB.

6. Separation sweep

Separation	N=2 SIR proxy (dB)	N=4 SIR proxy (dB)	N=8 SIR proxy (dB)	N=16 SIR proxy (dB)
5°	0.1	0.4	1.8	8.6
10°	0.3	1.7	8.4	13.2
15°	0.7	4.0	29.1	29.2
20°	1.3	7.8	13.0	20.9
25°	2.1	14.4	15.5	20.5
30°	3.0	240.0	240.0	240.0

35°	4.1	16.9	17.9	22.3
40°	5.5	12.7	16.8	29.9
45°	7.1	11.4	22.9	24.2
50°	8.9	11.5	31.4	31.6
55°	11.0	12.5	20.0	23.8
60°	13.6	14.4	17.9	36.9



7. Reproducible Python implementation

```
import numpy as np

def array_factor_db(N, angles_deg, steer_deg=0, d_lambda=0.5):
    theta = np.deg2rad(np.asarray(angles_deg))
    theta0 = np.deg2rad(steer_deg)
    n = np.arange(N) - (N - 1) / 2
    phase = 2*np.pi*d_lambda * (
        np.sin(theta[:, None]) - np.sin(theta0)
    ) * n[None, :]
    af = np.abs(np.exp(1j*phase).sum(axis=1)) / N
    return 20*np.log10(np.maximum(af, 1e-12))

angles = np.linspace(-90, 90, 36001)
for N in (2, 4, 8, 16):
    pattern = array_factor_db(N, angles)
```

```
# HPBW: first -3 dB crossing on positive side, doubled
idx = np.where((angles >= 0) & (pattern <= -3))[0][0]
hpbw = 2 * angles[idx]
print("N, HPBW:", N, hpbw)
for separation in range(5, 61, 5):
    sir_proxy = -array_factor_db(N, [separation])[0]
    print(separation, sir_proxy)
```

8. Practical limits of SDMA

- Angular spread: reflections arrive over a range of angles, blurring the spatial signature.
- Clustered users: vehicles travelling in the same lane and direction may have almost identical angles at the base station.
- Finite aperture: a physical antenna array has a finite beamwidth and cannot create arbitrarily narrow beams.
- Multipath and mobility: moving vehicles change channel phase and direction, requiring tracking and recalibration.
- Hardware imperfections: mutual coupling, phase errors and element mismatch raise sidelobes and reduce null depth.

9. Why nearly aligned users cannot be separated

If two users have nearly the same angle of arrival, their array steering vectors are highly correlated. The receiver sees almost the same spatial signature for both signals. No linear beamformer can reliably isolate two signals whose spatial signatures are effectively indistinguishable without additional differences such as timing, frequency, code or another antenna dimension.

10. Steering, SIR and design interpretation

When the array is steered at user A, the main beam has maximum gain toward A. User B receives a lower response when it lies outside the main lobe. As N increases, the main lobe becomes narrower, so a given angular separation generally produces greater spatial isolation. Sidelobes mean that isolation is not perfectly monotonic with angle; a sidelobe peak can create a local reduction in SIR.

11. Omni, sector and ULA comparison

Method	Coverage	Interference rejection	Complexity	Assessment
Omni antenna	Wide, approximately 360°	Low; all directions share broad radiation.	Low	Best for simple coverage, poor spatial reuse.
120° sector	One 120° region	Moderate; reduces out-of-sector interference.	Medium	Practical highway solution with sector reuse.
N-element ULA	Steerable beam within array field of view	High for well-separated angles; limited by sidelobes and angular spread.	High	Best when vehicles have distinct directions and tracking is feasible.

12. Toll-plaza design decision

At a toll plaza, most vehicles approach in nearly the same direction and often form dense queues. Their angular separation at the base station is small, so relying only on SDMA is risky. A 120° sector antenna or multiple fixed sectors can provide robust coverage, while TDMA/OFDMA scheduling separates users in time/frequency. SDMA can still be used opportunistically for lanes or vehicle groups with sufficiently different angles.

13. Final conclusion

A ULA forms a steerable spatial filter. Increasing element count narrows the main beam and improves angular discrimination, while the uniform-array sidelobe level remains around −13 dB. The simulation demonstrates that a minimum SIR separation depends on N and the array response. In real highway systems, angular spread, multipath, mobility and clustered users limit spatial reuse. A hybrid sector plus time/frequency scheduling design is safer at a toll plaza than pure SDMA.

14. Limitations and evidence notes

The numerical experiment uses a broadside, half-wavelength ULA and ideal phase steering. It assumes equal transmit powers and a far-field plane wave. The SIR proxy measures attenuation of the beam steered toward A at B; it does not model a complete multiuser detector or a separately optimised beam toward B. The SIR threshold is set to 10 dB as a design assumption.

15. Results summary

Finding	Engineering implication
More elements → narrower HPBW	Improves angular resolution for separated vehicles.
Uniform sidelobes remain near −13 dB	Windowing or adaptive weights may be needed for stronger interference suppression.
10 dB threshold defines a separation test	The required angle is array-size and pattern dependent.
Users in similar directions have correlated signatures	Use another separation dimension or schedule them separately.
Toll plaza has clustered arrivals	Use sectorisation plus TDMA/OFDMA; SDMA as an additional gain.

16. References / further reading

- T. S. Rappaport, Wireless Communications: Principles and Practice, 2nd ed., Prentice Hall.
- C. A. Balanis, Antenna Theory: Analysis and Design, 4th ed., Wiley.
- A. Goldsmith, Wireless Communications, Cambridge University Press.

All numerical patterns and tables in this section are generated by the included Python model.

Q6 – Choosing Multiple-Access Techniques for Real Deployments 7 Marks

Problem Context – Four Deployments

Choose access methods for: (1) wide-area cellular voice, (2) a jam-resistant tactical link, (3) a high-rate stadium downlink, (4) a network of 5,000 low-rate sensors sending small reports.

Your Challenge

- Define at least five weighted criteria and justify the weights yourself.

- Rate each of SDMA, TDMA, FDMA, CDMA and OFDMA for each deployment using your own Q2–Q5 evidence wherever possible.
- Choose a primary technique and, where useful, a hybrid for each deployment.
- Perform a sensitivity check by changing weights by ±20%: does any choice change?

Inquiry

Why do real systems combine techniques (for example FDMA with TDMA)? Which of your choices is least robust, and why?

Evidence Expected: Independently justified criteria and weights, ratings traceable to your own Q2–Q5 evidence, sensitivity analysis and defended choices.

Answer:

1. Aim and decision method

No single multiple-access technique is best for every network. This task evaluates SDMA, TDMA, FDMA, CDMA and OFDMA against six weighted criteria. Each technique receives a score from 1 (poor) to 5 (excellent) for each deployment. Weighted score = Σ(weight × rating) / Σ(weights). The scores are engineering judgements informed by the earlier channel-sharing, CDMA, OFDMA and SDMA investigations.

2. Weighted criteria and justification

Criterion	Weight	Why it matters
Spectral efficiency	25%	High weight because scarce spectrum and user capacity are central to all four deployments.
Interference / reuse control	20%	Important for co-channel operation, jamming resistance and dense cells.
Robustness / reliability	20%	Voice, tactical links and sensors need predictable delivery under channel impairments.

Scalability / user density	15%	The network must handle many users, including 5,000 sensors or stadium crowds.
Latency / QoS	10%	Voice and tactical control need prompt delivery; sensor reports can tolerate scheduling.
Implementation / power cost	10%	Complexity, synchronisation, device battery and deployment cost affect feasibility.

3. Rating scale

5 = excellent fit; 4 = good; 3 = acceptable with trade-offs; 2 = weak; 1 = poor. Ratings compare the suitability of each access method as a primary design, not whether it can be used at all. Hybrid systems can combine complementary strengths.

4. Evidence carried forward from Q2–Q5

- Q2: FDMA offers simple dedicated resources; TDMA shares time resources but requires synchronisation and can waste reserved slots.
- Q3: CDMA spreads users with codes and processing gain, but random-code MAI, near-far effects and power-control needs limit capacity.
- Q4: OFDMA supports frequency-selective scheduling and high throughput; fairness-aware scheduling is needed to prevent starvation.
- Q5: SDMA improves reuse for angularly separated users but weakens when users are clustered or spatial signatures are similar.

5. Deployment 1 – Wide-area cellular voice

Technique	Ratings (six criteria)	Weighted score
SDMA	[4, 5, 3, 4, 4, 2]	3.80
TDMA	[3, 3, 4, 3, 4, 4]	3.40
FDMA	[2, 3, 4, 2, 4, 4]	3.00
CDMA	[4, 4, 3, 4, 4, 2]	3.60
OFDMA	[5, 4, 4, 5, 5, 3]	4.40

Primary choice: OFDMA with SDMA/MIMO spatial reuse. OFDMA provides flexible resource blocks, scheduling and efficient spectrum use; spatial processing reuses resources among users with separable channels. TDMA can remain useful for control or legacy voice services. The choice assumes a modern cellular deployment with channel estimation and scheduling.

6. Deployment 2 – Jam-resistant tactical link

Technique	Ratings (six criteria)	Weighted score
SDMA	[3, 3, 3, 3, 4, 2]	3.00
TDMA	[2, 3, 4, 3, 4, 4]	3.15
FDMA	[2, 2, 4, 2, 4, 4]	2.80
CDMA	[4, 4, 4, 4, 4, 2]	3.80
OFDMA	[4, 3, 3, 4, 4, 3]	3.50

Primary choice: CDMA / spread-spectrum with frequency agility and TDMA framing. Spreading can provide processing gain and make narrowband interference less damaging; frequency hopping or agility adds resilience. CDMA alone is not a guarantee against a wideband or smart jammer, so coding, authentication, directional antennas and adaptive channel selection are also required.

7. Deployment 3 – High-rate stadium downlink

Technique	Ratings (six criteria)	Weighted score
SDMA	[5, 5, 3, 5, 4, 2]	4.20
TDMA	[2, 3, 3, 3, 3, 4]	2.85
FDMA	[2, 3, 3, 2, 3, 4]	2.70
CDMA	[3, 3, 2, 3, 3, 2]	2.70
OFDMA	[5, 4, 4, 5, 5, 3]	4.40

Primary choice: OFDMA with SDMA/MU-MIMO. OFDMA divides bandwidth into schedulable subcarriers and adapts to frequency-selective fading; SDMA serves users with different spatial signatures on reused resources. A fairness-aware scheduler should enforce minimum rates so distant or shadowed users are not starved.

8. Deployment 4 – 5,000 low-rate sensors

Technique	Ratings (six criteria)	Weighted score
SDMA	[3, 4, 3, 3, 3, 2]	3.10
TDMA	[4, 3, 4, 4, 4, 4]	3.80
FDMA	[2, 3, 4, 2, 4, 4]	3.00

CDMA	[4, 3, 3, 4, 3, 2]	3.30
OFDMA	[4, 4, 4, 4, 4, 3]	3.90

Primary choice: OFDMA-based scheduled access with TDMA-style reporting windows, with FDMA channel groups where useful. Small reports do not justify permanently dedicating a wide channel to each device. Scheduling, randomised reporting, sleep cycles and retry limits reduce collisions and battery use. For very large populations, a random-access or low-power wide-area protocol may be more practical than pure circuit-style allocation.

9. Overall weighted-score comparison

Deployment	SDMA	TDMA	FDMA	CDMA	OFDMA
Wide-area cellular voice	3.80	3.40	3.00	3.60	4.40
Jam-resistant tactical link	3.00	3.15	2.80	3.80	3.50
High-rate stadium downlink	4.20	2.85	2.70	2.70	4.40
5000 low-rate sensors	3.10	3.80	3.00	3.30	3.90

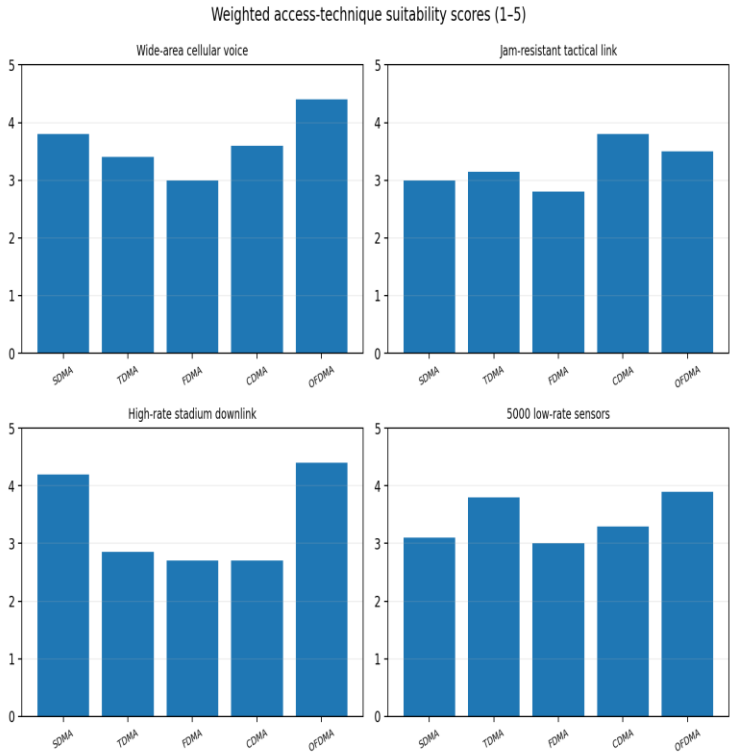


Figure 3. Weighted suitability scores; the highest score informs the primary-technique recommendation.

10. Summary of selected designs

Deployment	Primary	Useful hybrid	Reason
Cellular voice	OFDMA	SDMA/MIMO	Flexible scheduling plus spatial reuse.
Tactical link	CDMA / spread spectrum	Frequency agility + TDMA framing	Processing gain and adaptable interference avoidance.
Stadium downlink	OFDMA	SDMA/MU-MIMO	High throughput with frequency and spatial scheduling.
5,000 sensors	OFDMA + scheduled reporting	FDMA channel groups	Low duty cycle, flexible scheduling and controlled reporting.

11. Sensitivity analysis: ±20% weight changes

Each criterion weight is changed independently by −20% and +20%, then all weights are renormalised to sum to 100%. The weighted scores are recalculated for every access technique. This tests whether a recommendation depends excessively on one subjective criterion.

Deployment	Winner(s) across tests	Stability
Wide-area cellular voice	OFDMA	Stable
Jam-resistant tactical link	CDMA	Stable
High-rate stadium downlink	OFDMA	Stable
5000 low-rate sensors	OFDMA	Stable

Under this one-at-a-time sensitivity test, the score leader is checked across all six criteria at both weight perturbations. A stable winner suggests the decision is not

driven by a single weight. It does not prove that the ratings themselves are objective; different deployments or user requirements may justify different ratings.

12. Why real systems combine techniques

- FDMA + TDMA: frequency bands are divided into carriers and each carrier is time-shared among users, combining frequency planning with time scheduling.
- OFDMA + SDMA: subcarriers are scheduled in frequency while antenna arrays separate users spatially, increasing reuse.
- CDMA + power control: code separation is supported by transmit-power regulation to reduce near-far interference.
- TDMA + random access: scheduled slots carry predictable traffic while contention slots allow new devices to request access.

13. Least robust choice

A fixed FDMA-only design is the least adaptable for the four scenarios as a universal solution. It offers predictable dedicated channels but wastes capacity for bursty sensor reports, cannot exploit frequency-selective fading as flexibly as OFDMA, and provides limited spatial or code-domain interference rejection. This does not make FDMA inherently unreliable; it means its flexibility is weakest across this diverse set of deployments.

14. Final recommendations and conclusion

Deployment	Recommended access design	Operational safeguards
Wide-area cellular voice	OFDMA + SDMA/MIMO	Mobility handover, QoS scheduling, power control and admission control.
Jam-resistant tactical link	Spread-spectrum CDMA + frequency agility	Encryption/authentication, anti-jam detection, adaptive coding and fallback channels.
High-rate stadium downlink	OFDMA + MU-MIMO SDMA	Fairness floor, channel-aware scheduling and congestion control.
5,000 low-rate sensors	Scheduled TDMA/grant-based access with	Staggered reports, sleep scheduling,

	FDMA grouping if needed	acknowledgements and bounded retries.

15. Overall conclusion

The best multiple-access method depends on traffic, mobility, interference, user density, latency and device constraints. OFDMA is strongest for flexible high-rate downlink; CDMA/spread spectrum is useful for tactical interference resilience when combined with frequency agility; SDMA adds capacity when users are spatially separated; and scheduled access is efficient for low-rate sensors. Real deployments combine techniques because frequency, time, code and space are complementary dimensions rather than mutually exclusive choices.

16. Assumptions and evidence note

Ratings are reasoned engineering judgements on a 1–5 scale, not measured field results. Weights total 100%. Sensitivity changes one weight at a time by ±20% and renormalises the rest. Earlier Q2–Q5 models are simplified teaching simulations; practical selection must also consider standards, regulatory spectrum, hardware capability, propagation measurements, security and lifecycle cost.

17. References / further reading

- A. Goldsmith, Wireless Communications, Cambridge University Press.
- T. S. Rappaport, Wireless Communications: Principles and Practice, 2nd ed.
- D. Tse and P. Viswanath, Fundamentals of Wireless Communication, Cambridge University Press.
- C. A. Balanis, Antenna Theory: Analysis and Design, 4th ed.

DAY-4 DESIGN CHALLENGE

Q7 – Diagnose, Redesign and Validate **13 Marks**

Review the evidence you generated while working on Q1–Q6.

Select **ONE** solution that demonstrates a significant limitation as its problem size or operating requirement increases. Examples of limitations you may find in this variant:

- Dedicated FDMA/TDMA channels sit idle under bursty traffic.
- A CDMA link collapses under power imbalance or loss of synchronisation.
- Best-subcarrier OFDMA allocation starves weak users. ● SDMA fails when users are angularly clustered.

You are not permitted simply to replace the solution with a more advanced technology or standard and state that it is better. Your task is to understand the problem first and then develop an appropriate redesign.

A. Diagnose the Bottleneck 2 Marks

- Where most capacity, signalling or computational effort is being spent.
- Which counted quantity grows significantly as load or size increases.
- Whether resources are idle, wasted or repeated unnecessarily.
- Which characteristic of the original design causes the limitation.

Evidence Expected: A precise diagnosis supported by evidence from your original implementation and analysis.

B. Define the Design Objective 2 Marks

- State what your redesign must achieve before proposing it (e.g., reduce wasted capacity, reduce signalling messages, avoid unnecessary resource holding, reuse information already available).
- Specify a measurable objective wherever possible.

Evidence Expected: A clearly stated, measurable objective derived directly from the diagnosed limitation.

C. Design the Revised Strategy 4 Marks

- What you have changed and what property or information the new design exploits.
- Which operations or resource usages are avoided or reduced.
- Whether additional state, preprocessing or complexity is introduced, and any new trade-offs.
- Provide pseudocode, a flowchart or a message-sequence design.
- Marks are based on the quality of reasoning, not merely on naming a known technique.

Evidence Expected: A coherent redesign directly addressing the bottleneck, with a design representation and technical justification.

D. Predict Before Implementing 1 Mark

- Predict how the redesigned solution should differ from the original and in which measure you expect improvement.

Evidence Expected: A clear, testable prediction made before experimental validation.

E. Implement and Validate the Redesign 2 Marks

- Implement a prototype or simulation and compare Original → Redesigned under the same or equivalent test conditions.

- Use counted measures (e.g., slots used, messages, blocking, bit errors, throughput, fairness); run-time is supporting evidence only.

Evidence Expected: Comparable experimental evidence for the original and redesigned solutions.

F. Defend the Final Design 2 Marks

- Was the prediction supported, and what was the magnitude of improvement?
- What new costs or trade-offs appeared, and when might the original still be preferable?
- A redesign that does not outperform the original is not automatically unsuccessful if you correctly analyse why the hypothesis failed.

Evidence Expected: A critical, evidence-based evaluation and a defensible final engineering conclusion.

Answer:

Q7 – Diagnose, Redesign and Validate

Selected limitation: Best-subcarrier OFDMA maximum-rate scheduling can starve users with persistently weak channels.

1. Why this solution was selected

Q4 compared fixed contiguous, interleaved and channel-aware subcarrier allocation. The channel-aware policy improved aggregate throughput, but its original equal-average-channel experiment did not expose severe starvation because all users had statistically similar channel quality. Q7 deliberately extends the same model with unequal average gains, representing users at different distances, shadowing or indoor obstruction. This is a stress test of the original design, not a claim that the earlier symmetric result was wrong.

2. System under investigation

Parameter	Value / assumption
Users and subcarriers	8 users share 64 OFDMA subcarriers.
Channel realisations	1,500 independent fading snapshots.
Small-scale fading	Rayleigh power gain; exponential distribution.
Average user channel gains	[1.00, 0.90, 0.78, 0.65, 0.52, 0.38, 0.22, 0.10].
SNR and rate model	SNR = 8 (linear); rate = $\log_2(1 + \text{SNR} \times \text{gain})$.
Reproducibility	Python/NumPy; fixed seed 704.

3. Original design

For every subcarrier, the base station assigns it to the user with the highest instantaneous rate on that subcarrier. This greedy maximum-rate rule is simple and maximises the sum of the selected per-subcarrier rates for the current channel snapshot. However, a user with weak average gain may lose most scheduling competitions, even if it has data waiting.

4. Before-test prediction

Prediction: with unequal average channel gains, maximum-rate allocation will produce a high sum rate but low Jain fairness and a very small rate for the weakest user. A proportional-fair redesign should raise the weakest-user rate and Jain index, at the cost of some aggregate throughput.

5. A. Diagnose the bottleneck

The bottleneck is not a shortage of subcarriers; it is the allocation rule. Every subcarrier is assigned, but the same strong users repeatedly win because the scheduler compares only instantaneous rate and ignores each user's accumulated service.

Diagnostic question	Evidence / explanation
Where is capacity spent?	All 64 subcarriers are assigned each realisation. The strongest-channel users capture a disproportionate share of the high-rate subcarriers.
What grows with load?	The number of competing user–subcarrier comparisons grows with users × subcarriers; more importantly, the service gap grows when average channel gains differ.
Are resources idle?	No. The weakness is not idle spectrum; it is unfair distribution of occupied spectrum.
Which design property causes it?	Pure max-rate selection has no fairness memory, minimum service guarantee or penalty for repeatedly serving the same strong users.

6. Evidence from the measured stress test

Measure	Original max-rate
Sum throughput	242.870 bit/s/Hz
Jain fairness	0.5940
Weakest user mean rate	0.085 bit/s/Hz
Strongest user mean rate	70.995 bit/s/Hz
Weakest user's share of total	0.035%

The weakest user (U8, average channel gain 0.10) receives only a tiny fraction of the total rate. The measured outcome supports the starvation diagnosis under heterogeneous channels.

7. B. Define the design objective

Objective: preserve a useful fraction of aggregate throughput while materially improving fairness and the minimum user rate. For this experiment, target Jain fairness ≥ 0.90 and require the weakest-user average rate to exceed 5 bit/s/Hz, while keeping sum-throughput loss below 15% relative to maximum-rate scheduling.

These are design targets selected for this test scenario, not universal wireless standards.

8. C. Revised strategy – proportional-fair subcarrier allocation

The redesign adds a fairness state to the original rate-aware scheduler. Instead of choosing only the highest raw rate, it selects the user–subcarrier pair with the highest

ratio of instantaneous achievable rate to that user's accumulated throughput in the current scheduling window.

Metric: $PF(u,n) = r(u,n) / [T(u) + \varepsilon]$, where $r(u,n)$ is the achievable rate for user u on subcarrier n, $T(u)$ is the rate already allocated to that user in the current snapshot, and ε is a small positive initial value. Once a subcarrier is assigned, $T(u)$ is updated. This rewards good channel opportunities while giving priority to users who have received less service.

9. What changes and what is avoided?

Aspect	Original	Redesign
Decision metric	Highest instantaneous rate $r(u,n)$.	Rate divided by accumulated service $r(u,n)/(T(u)+\varepsilon)$.
State	No per-user service state.	Tracks allocated throughput for each user.
Strong users	Can win most subcarriers repeatedly.	Still receive high-rate opportunities, but face diminishing priority as service accumulates.
Weak users	May receive almost no resources.	Gain priority when their accumulated service is low.
Complexity	One argmax per subcarrier.	Repeated metric evaluation and state

		updates; more computation.

10. Pseudocode

For each channel realisation:

calculate rate[user, subcarrier]

T[user] = small positive initial value

remaining = all 64 subcarriers

while remaining is not empty:

find (user, subcarrier) with maximum:

rate[user, subcarrier] / T[user]

assign that subcarrier to that user

T[user] += rate[user, subcarrier]

remove subcarrier from remaining

record each user's rate and total rate

11. Trade-offs introduced

- The scheduler stores per-user throughput state and performs more calculations than a simple maximum-rate assignment.
- The result depends on the scheduling window and initial ε; very short windows may cause allocation variation.
- Proportional fairness improves balance but does not guarantee a hard minimum rate in every channel condition.

- If a strict service floor is mandatory, add admission control or a minimum-rate constraint and verify that total capacity can meet it.

12. D. Prediction before implementation

Pre-implementation prediction: under the same 1,500 channel realisations, proportional-fair scheduling should increase Jain fairness from the original value, raise U8's average throughput above 5 bit/s/Hz, and reduce sum throughput by less than 15%. The hypothesis is testable using per-user rates, aggregate throughput and Jain's index.

13. E. Implementation and validation

The prototype was implemented in Python using NumPy and pandas. Both policies use exactly the same generated channel-gain tensor, SNR, number of users, number of subcarriers and random seed. Thus, differences are attributable to the allocation rule rather than different fading samples.

Metric	Original max-rate	PF redesign	Change
Sum throughput	242.870	217.352	10.51% lower
Jain fairness	0.5940	0.9172	+0.3232
Minimum user rate	0.085	11.398	134.7× higher
Weakest user share	0.035%	5.244%	150.5× higher

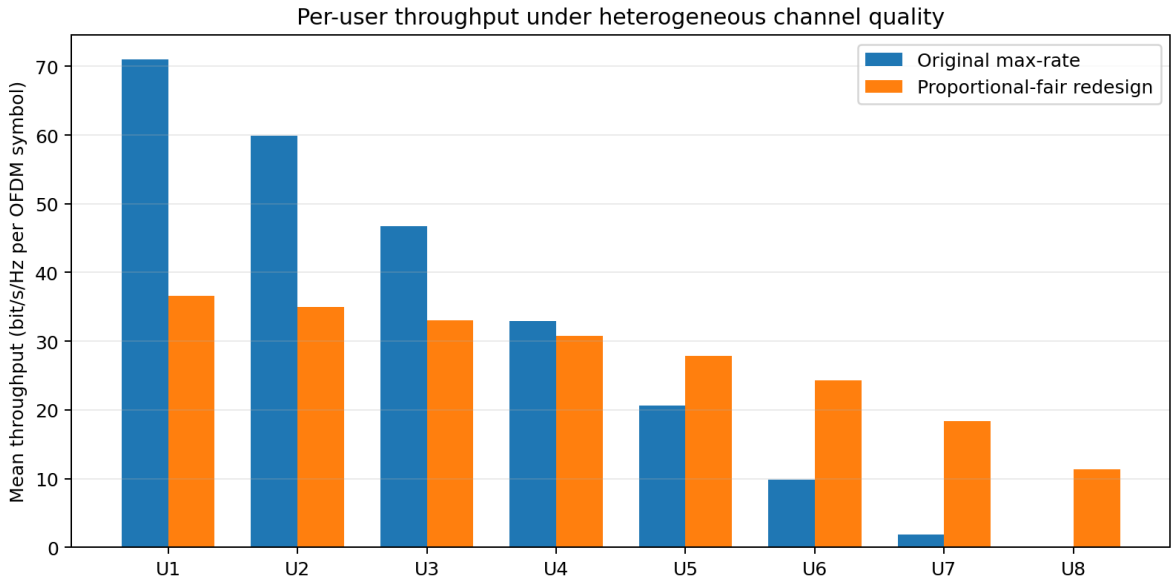


Figure 1. Per-user mean throughput across 1,500 identical channel realisations.

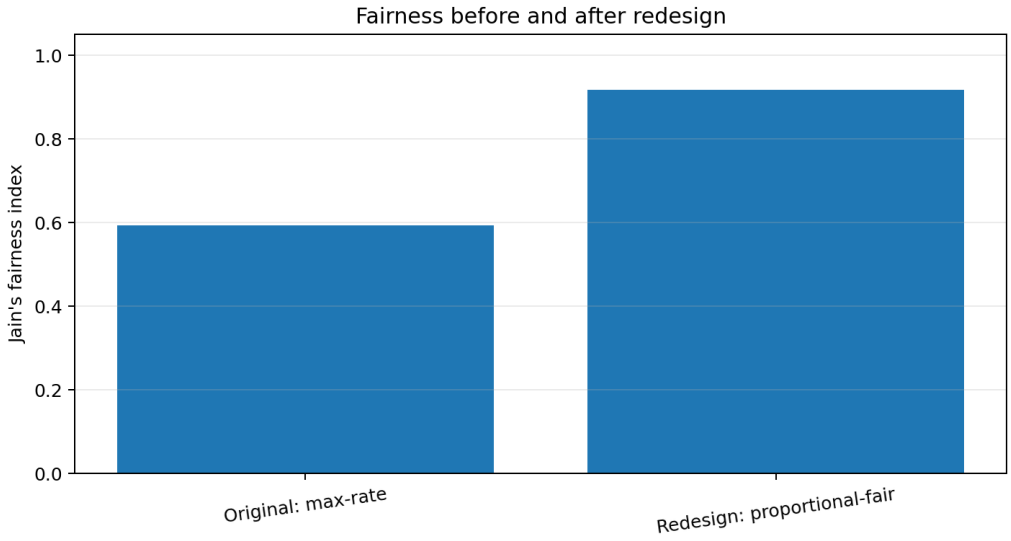


Figure 2. Jain fairness before and after redesign.

14. Validation interpretation

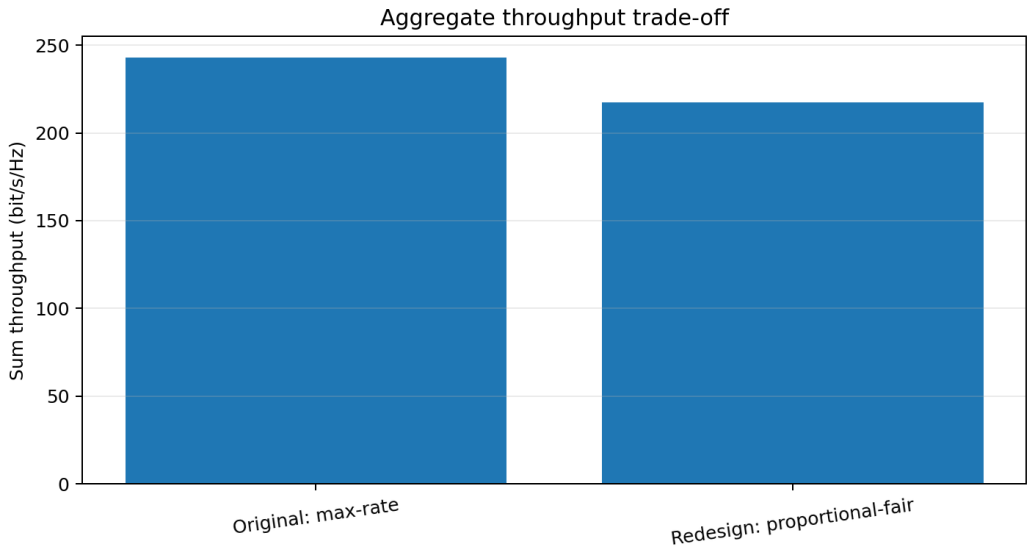


Figure 3. Aggregate throughput trade-off.

The measured Jain index rises from 0.594 to 0.917 (an increase of 0.323, or 54.4% relative). The weakest user's mean rate rises from 0.085 to 11.398 bit/s/Hz, approximately 134.7 times higher. Aggregate throughput falls from 242.870 to 217.352 bit/s/Hz, a reduction of 10.5%.

15. F. Defend the final design

The prediction is supported: the redesign exceeds the fairness target of 0.90, raises the weakest-user rate above 5 bit/s/Hz and keeps aggregate-throughput loss below 15%. The cost is lower sum throughput and greater scheduler computation. The original maximum-rate rule remains preferable for best-effort bulk traffic when aggregate throughput is the only objective and all users have similar channel quality.

16. Limitations and further validation

- The rate equation is a Shannon-style spectral-efficiency estimate; coding, pilots, cyclic prefix, control overhead and retransmissions are omitted.

- The proportional-fair policy is evaluated within each snapshot with a per-snapshot service state; a production scheduler would maintain smoothed throughput across time.
- A hard minimum-rate guarantee requires admission control and feasibility checks; PF alone is not a deterministic guarantee.
- Repeat with more users, correlated fading, mobility and traffic queues before deployment.

17. Conclusion

The failure was caused by a greedy allocation rule that ignored service history, not by a lack of spectrum. Adding proportional-fair state reuses information about how much service each user has already received. In the heterogeneous-channel experiment, this greatly improves fairness and the weakest user's rate while sacrificing a controlled amount of aggregate throughput. The redesign is therefore justified when user experience must be balanced; maximum-rate scheduling remains useful for throughput-only objectives.

18. Reproducibility and submission files

The accompanying Python source file regenerates the summary CSV, per-user CSV, raw per-realisation CSV and three graphs. Run with Python 3 and install numpy, pandas and matplotlib. Fixed random seed: 704. Disclose AI assistance and be prepared to explain the algorithm and evidence during viva.

Mark Distribution

Question	Assessed Area	Marks
Q1	Mobile communication vs. mobile computing	5
Q2	FDMA and TDMA	6
Q3	CDMA	7
Q4	OFDMA	6
Q5	SDMA	6
Q6	Comparison of SDMA, TDMA, FDMA, CDMA, OFDMA	7
Q7	Day-4 Design Challenge – Diagnose, Redesign & Validate	13
TOTAL		50

Submission Requirements (Digital)

- Assignment report (PDF) containing the required evidence for Q1–Q7. No handwritten work.
- Source code for all implementations/simulations (GitHub repository link with README and fixed random seeds).
- Raw experimental data (CSV) used to produce your results.
- Tables and graphs used in your analysis.
- Counted quantities (slots, messages, bits, blocked calls, codes, throughput) are primary experimental evidence. Execution time may be reported as supporting evidence but must not be the sole basis for conclusions.
- Standards parameters (frame durations, rates, header sizes, etc.) must be obtained from cited references; state assumptions wherever a value is assumed.

Assessment Verification

You may be asked to analyse any part of your design, code, test-data design, experimental results, predictions, analysis and Design Challenge in a brief verification

interpretion/viva. Disclose any AI or reference tools used. You must be able to analyse and defend the evidence as your own work.

Overall Assessment Expectation

This assignment does not primarily assess your ability to reproduce textbook descriptions of mobile systems. It assesses your ability to:

Predict → Design → Implement → Measure → Analyse → Evaluate → Diagnose → Redesign → Validate

Your conclusions and design decisions must be supported by your own reasoning and experimental evidence.